

tower property

$$E \left(\underbrace{E(f(X_1 \dots X_n) \mid X_1 \dots X_k)}_{\text{tower property}} \mid X_1 \dots X_{k-1} \right) \\ = E(f(X_1 \dots X_n) \mid X_1 \dots X_{k-1})$$

Martingale

$$(Y_k, \mathcal{F}_k)_{k=1}^n$$

$$① E(|Y_k|) < \infty$$

$$② E(Y_{k+1} \mid \underbrace{\mathcal{F}_k}_{\uparrow}) = Y_k$$

$\mathcal{F}_k$ filtration

$$\underbrace{\mathcal{F}_1}_{=} \subseteq \mathcal{F}_2 \subseteq \dots$$

$$\mathcal{F}_k = \sigma(X_1 \dots X_k)$$

Ex. ① $X_1, \dots, X_n$ iid μ $S_k = \sum_{i=1}^k X_i$ $\{S_k, \sigma(X_1, \dots, X_k)\}_{k=1}^n$

is it martingale?

$$E(|S_k|) \leq k \underbrace{E(|X_i|)}_{< \infty}$$

$$E(S_{k+1} | X_1, \dots, X_k) \stackrel{?}{=} S_k$$

$$= E(S_k + X_{k+1} | X_1, \dots, X_k)$$

$$= E(S_k | X_1, \dots, X_k) + E(X_{k+1} | X_1, \dots, X_k)$$

if $\mu = 0$

$$= S_k + \underbrace{E(X_{k+1} | X_1, \dots, X_k)}_{\substack{\uparrow \\ \text{ind.} \\ E(X_{k+1})}}$$

Ex. 2

$$Y_k = E(f(X_1, \dots, X_n) | X_1, \dots, X_k) \quad \text{Doob's Construction}$$

$\{Y_k, \sigma(X_1, \dots, X_k)\}_{k=1}^n$ is it martingale?

$$\forall k \quad E(|Y_k|) = E(|E(f(X_1, \dots, X_n) | X_1, \dots, X_k)|)$$

$$\leq E(E(|f(X_1, \dots, X_n)| | X_1, \dots, X_k)) = E(|f(X_1, \dots, X_n)|) < \infty$$

Similarly,

$$E_x(E(Y | X)) = E(Y)$$

Assumption ①

$$E(Y_{k+1} | X_1, \dots, X_k) \stackrel{?}{=} Y_k$$

tower prop.

$$E(E(f(X_1, \dots, X_n) | X_1, \dots, X_{k+1}) | X_1, \dots, X_k) \stackrel{?}{=} E(f(X_1, \dots, X_n) | X_1, \dots, X_k) = Y_k$$

$$\underline{Y_n - Y_0} = \underbrace{(Y_n - Y_{n-1})}_{D_n} + \underbrace{(Y_{n-1} - Y_{n-2})}_{D_{n-1}} + \dots$$

if $\{Y_k, \sigma(X_1, \dots, X_k)\}_{k=1}^n$ is martingale then

$D_k = Y_k - Y_{k-1}$ is called martingale diff. seq. $\nearrow$

Ex. 3 Likelihood Ratio

f, g density

$X_1, \dots, X_n$ iid $\underline{f}$

absol. Cont.

$$Y_n = \prod_{l=1}^n \frac{g(X_l)}{f(X_l)}$$

$$E(|Y_n|) = E\left(\prod_{l=1}^n \frac{g(X_l)}{f(X_l)}\right) = \prod_{l=1}^n E\left(\frac{g(X_l)}{f(X_l)}\right) \overset{\text{absol. Cont.}}{=} \int_{-\infty}^{+\infty} \frac{g(x)}{f(x)} \cdot \cancel{f(x)} \cdot dx = \int_{-\infty}^{+\infty} g(x) dx = 1$$

$$\textcircled{2} \quad \mathbb{E}(Y_{k+1} \mid X_1, \dots, X_k) \stackrel{?}{=} Y_k$$

$$\mathbb{E}\left(\prod_{l=1}^{k+1} \frac{g(X_l)}{f(X_l)} \mid X_1, \dots, X_k\right) = \underbrace{\prod_{l=1}^k \frac{g(X_l)}{f(X_l)}}_{Y_k} \mathbb{E}\left(\frac{g(X_{k+1})}{f(X_{k+1})} \mid X_1, \dots, X_k\right)$$

Th. 2.19

Suppose $\{(D_k, \mathcal{F}_k)\}_{k=1}^{\infty}$

be a martingale diff. Seq.

$$\Rightarrow \mathbb{E}\left(e^{\lambda D_k} \mid \mathcal{F}_{k-1}\right) \leq e^{\lambda^2 v_k^2 / 2} \quad \text{for } |\lambda| < \frac{1}{\alpha_k} \quad \text{almost surely.}$$

Then, $\sum_{k=1}^n D_k$ is

sub-exp.

$$\left(\sqrt{\sum_{k=1}^n v_k^2}, \alpha_x\right) \text{ s.t.}$$

$$\alpha_x = \max_{i=1 \dots n} \alpha_i.$$

$$\begin{aligned}
 \mathbb{E} \left\{ e^{\lambda \sum_{k=1}^n D_k} \right\} &= \mathbb{E} \left(\mathbb{E} \left(e^{\lambda D_n} \cdot \underbrace{e^{\lambda \sum_{k=1}^{n-1} D_k}}_{\mathcal{F}_{n-1}} \middle| \mathcal{F}_{n-1} \right) \right) \\
 &= \mathbb{E} \left(e^{\lambda \sum_{k=1}^{n-1} D_k} \underbrace{\mathbb{E} \left(e^{\lambda D_n} \middle| \mathcal{F}_{n-1} \right)}_{\lambda^2 v_n^2 / 2} \right) \\
 &\leq \underbrace{\mathbb{E} \left(e^{\lambda \sum_{k=1}^{n-1} D_k} \right)}_{\prod_{i=1}^n e^{\lambda^2 v_i^2 / 2}} \cdot e^{\lambda^2 v_n^2 / 2} \quad |\lambda| < \frac{1}{\alpha_n} \\
 &\vdots \\
 &\leq \prod_{i=1}^n e^{\lambda^2 v_i^2 / 2} \quad |\lambda| < \frac{1}{\max_{i=1, \dots, n} \alpha_i} \\
 &= e^{\lambda^2 \underbrace{\left(\sum_{i=1}^n v_i^2 \right)}_{v^{*2} / 2} / 2} \quad v^* = \sqrt{\sum_{i=1}^n v_i^2}
 \end{aligned}$$

$$\mathbb{E}(\mathbb{E}(Y|X)) = \mathbb{E}(Y)$$

if $X \sim \text{Sub-exp}(\nu, \alpha)$

$$P(|X - \underline{\underline{E(X)}}| > t) \leq \begin{cases} 2e^{-\frac{t^2}{2\nu^2\alpha}} & 0 \leq t \leq \frac{\nu^2}{\alpha} \\ 2e^{-\frac{t}{2\alpha}} & t > \frac{\nu^2}{\alpha} \end{cases}$$

$$X \leftarrow \sum_{k=1}^n D_k$$

$$E(X) = \sum \widehat{E(D_k)}$$

$$E(D_k) = E(Y_{k+1} - Y_k)$$

$$= E(E(f(X_1, \dots, X_n) | X_1, \dots, X_{k+1}))$$

$$- E(f(X_1, \dots, X_n) | X_1, \dots, X_k) = 0$$

$$P\left(\left|\sum_{k=1}^n D_k\right| > t\right) \leq \begin{cases} \nu = \nu^* = \sqrt{\sum_{i=1}^n \nu_i^2} \end{cases}$$